

# Mutual Fund Analytics

A Comprehensive Data Analytics and Business Intelligence Capstone Project

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Submitted By

**HARSHITHA SUNIL**

**Internship : Bluestock Fintech**

**Submitted To : Bluestock Fintech**

**Project Duration : June 2026 – July 2026**

**Technologies Used :**

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- SQLite
- SQL
- Power BI
- Git
- GitHub

## Certificate

This is to certify that the project entitled "**Mutual Fund Analytics – A Comprehensive Data Analytics and Business Intelligence Capstone Project**" has been successfully completed by **Harshitha Sunil**, during the internship at **Bluestock Fintech**.

The project was carried out as part of the internship curriculum and demonstrates the practical application of Data Analytics, Python Programming, SQL, SQLite, Financial Analytics, and Business Intelligence using Power BI.

The work presented in this report is original and has been completed under the guidance provided during the internship. To the best of our knowledge, this work has not been submitted previously for any other academic or professional purpose.

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## Acknowledgement

I would like to express my sincere gratitude to **Bluestock Fintech** for providing me with the opportunity to work on this capstone project. This internship has provided me with valuable practical exposure to real-world financial data analytics, database management, business intelligence, and data visualization.

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## Abstract

The **Mutual Fund Analytics** project focuses on analysing mutual fund performance, investor behaviour, and industry trends using Data Analytics and Business Intelligence techniques. The project integrates multiple financial datasets containing Net Asset Value (NAV) history, investor transactions, fund performance metrics, SIP inflows, portfolio holdings, benchmark indices, and market information.

The project follows an end-to-end analytics pipeline beginning with data ingestion, data cleaning, preprocessing, database creation using SQLite, exploratory data analysis, performance analytics, advanced financial analytics, and interactive dashboard development using Power BI.

Several financial performance indicators including Compound Annual Growth Rate (CAGR), Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Fund Scorecard, and Value at Risk (VaR) were calculated to evaluate mutual fund performance. Additional analytics such as Cohort Analysis and a Fund Recommendation System were implemented to provide meaningful investment insights.

The final outcome of the project is an interactive Power BI dashboard consisting of four analytical pages that enable users to analyse industry trends, fund performance, investor behaviour, and SIP trends. The project demonstrates practical implementation of data engineering, financial analytics, and business intelligence techniques to support data-driven investment analysis and decision-making.

# 1. Introduction

The mutual fund industry has become one of the most preferred investment avenues for both retail and institutional investors. With increasing participation in financial markets, investors require reliable analytical tools to evaluate fund performance, understand market trends, and make informed investment decisions.

The objective of this project is to build a comprehensive Mutual Fund Analytics system capable of analysing financial datasets using Data Analytics, SQL, Python, and Business Intelligence techniques. The project processes historical NAV data, investor transactions, portfolio holdings, and benchmark information to generate meaningful insights regarding fund performance and investor behaviour.

The project also demonstrates the complete data analytics lifecycle, including data ingestion, preprocessing, exploratory analysis, financial performance evaluation, advanced analytics, and dashboard development. The final interactive Power BI dashboard enables users to explore key performance indicators, compare mutual funds, analyse investor demographics, and monitor SIP trends through dynamic visualizations.

## 2. Problem Statement

The mutual fund industry generates a large volume of financial data, including Net Asset Value (NAV), investor transactions, portfolio holdings, benchmark indices, SIP inflows, and fund performance metrics. Analysing these datasets manually is time-consuming and makes it difficult to identify investment opportunities, compare mutual fund performance, and understand investor behaviour.

Traditional reporting methods often lack interactive visualization and advanced analytical capabilities. Investors and financial analysts require an integrated system that can clean, process, analyse, and visualize mutual fund data efficiently.

This project addresses these challenges by developing an end-to-end Mutual Fund Analytics platform that combines data engineering, financial analytics, SQL database management, and interactive dashboards to generate meaningful insights for investment decision-making.

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## 3. Objectives

The primary objectives of this project are:

- To collect and integrate multiple mutual fund datasets from different sources.
- To perform data cleaning and preprocessing for accurate analysis.
- To design and implement a SQLite database for efficient data storage and querying.
- To perform Exploratory Data Analysis (EDA) to understand industry trends and investor behaviour.
- To calculate key financial performance metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, and Fund Scorecard.
- To implement advanced analytics techniques including Value at Risk (VaR), Cohort Analysis, and a Fund Recommendation System.
- To develop an interactive Power BI dashboard for visualizing industry insights and fund performance.
- To demonstrate a complete end-to-end financial analytics pipeline using Python, SQL, SQLite, Git, GitHub, and Power BI.

## 4. Scope of the Project

The scope of this project includes the complete analytics lifecycle of mutual fund data.

The project covers:

- Data ingestion from multiple mutual fund datasets.
- Data cleaning, preprocessing, and validation.
- Database creation using SQLite.
- Exploratory Data Analysis (EDA).
- Performance analytics using financial metrics.
- Advanced analytics including Value at Risk (VaR), Cohort Analysis, and Fund Recommendation.
- Interactive dashboard development using Power BI.
- Version control using Git and GitHub.
- Documentation and reporting.

The project is designed as an educational capstone and demonstrates practical implementation of data analytics techniques on financial datasets.

## 5.Tools and Technologies Used

| Category             | Technology                |
|----------------------|---------------------------|
| Programming Language | Python                    |
| Data Processing      | Pandas, NumPy             |
| Data Visualization   | Matplotlib, Seaborn       |
| Database             | SQLite                    |
| Query Language       | SQL                       |
| Dashboard            | Power BI                  |
| Version Control      | Git                       |
| Repository Hosting   | GitHub                    |
| IDE                  | Jupyter Notebook, VS Code |
| Operating System     | Windows 11                |

## 6. Project Architecture

The project follows an end-to-end data analytics workflow consisting of the following stages:

### Step 1 – Data Collection

- Mutual fund datasets were collected from the provided Bluestock datasets.

### Step 2 – Data Cleaning

- Missing values were handled.
- Duplicate records were removed.
- Data types were corrected.
- Invalid values were cleaned.

### Step 3 – Database Design

- Cleaned datasets were stored in SQLite.
- Relationships were created using **AMFI Code** as the primary key.

### Step 4 – Exploratory Data Analysis

- Industry trends were analysed.
- Investor demographics were studied.
- SIP trends were visualized.
- Portfolio holdings were examined.

### Step 5 – Performance Analytics

- Daily Returns
- CAGR
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown
- Fund Scorecard



### **Step 6 – Advanced Analytics**

- Historical Value at Risk (VaR)
- Cohort Analysis
- Fund Recommendation System

### **Step 7 – Dashboard Development**

- Industry Overview
- Fund Performance
- Investor Analytics
- SIP & Market Trends

## 7. Dataset Description

The project uses multiple datasets representing different aspects of the mutual fund industry.

| Dataset               | Description                                         |
|-----------------------|-----------------------------------------------------|
| Fund Master           | Basic information about mutual fund schemes         |
| NAV History           | Historical daily Net Asset Value data               |
| Investor Transactions | Purchase, SIP, Redemption, and Lumpsum transactions |
| Monthly SIP Inflows   | Industry SIP investment data                        |
| Scheme Performance    | Fund performance metrics                            |
| Portfolio Holdings    | Stocks held by each mutual fund                     |
| Category Inflows      | Net inflow by mutual fund category                  |
| Benchmark Indices     | Nifty benchmark data for comparison                 |

## 8. ETL Pipeline (Extract, Transform and Load)

### 8.1 Overview

The ETL (Extract, Transform, and Load) pipeline was implemented to collect, clean, validate, and prepare mutual fund datasets for analysis. The pipeline ensures that the data is accurate, consistent, and suitable for financial analytics and dashboard development.

The ETL process was developed using Python and Pandas, enabling efficient handling of large datasets containing historical NAV values, investor transactions, portfolio holdings, benchmark indices, SIP inflows, and fund performance metrics.

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### 8.2 Extract Phase

The extraction phase involved importing multiple CSV datasets into Python using the Pandas library.

The datasets used include:

- Fund Master
- NAV History
- Investor Transactions
- Monthly SIP Inflows
- Scheme Performance
- Portfolio Holdings
- Category Inflows
- Benchmark Indices

These datasets were stored in the **data/raw/** directory before preprocessing.

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### 8.3 Transform Phase

During the transformation phase, several preprocessing techniques were applied to improve data quality.

The following operations were performed:

- Removal of duplicate records
- Handling of missing values
- Correction of inconsistent data types
- Date format standardization
- Validation of AMFI codes
- Feature engineering for financial metrics
- Creation of derived columns such as Daily Return

The cleaned datasets were stored in the **data/processed/** directory for further analysis.

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#### 8.4 Load Phase

After preprocessing, the cleaned datasets were loaded into a SQLite database.

Each dataset was imported into its respective table while maintaining relationships using **AMFI Code** as the primary linking attribute.

The database served as the central repository for querying and dashboard development.

## 9. SQLite Database Design

### 9.1 Database Overview

SQLite was selected as the project database because it is lightweight, portable, and easy to integrate with Python and Power BI.

The database stores all cleaned datasets in structured tables, enabling efficient querying and data retrieval.

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### 9.2 Database Tables

The database contains the following tables:

| Table Name            | Purpose                        |
|-----------------------|--------------------------------|
| Fund Master           | Stores scheme details          |
| NAV History           | Stores daily NAV values        |
| Investor Transactions | Stores investment transactions |
| Monthly SIP Inflows   | Stores SIP trend information   |
| Scheme Performance    | Stores performance metrics     |
| Portfolio Holdings    | Stores portfolio allocation    |
| Category Inflows      | Stores category-wise inflows   |
| Benchmark Indices     | Stores benchmark market data   |

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### 9.3 Database Relationships

The database was designed using **AMFI Code** as the primary relationship key.

Relationships allow integration of:

- NAV History with Scheme Performance
- Portfolio Holdings with Fund Master
- Investor Transactions with Fund Master

This relational structure enables efficient analysis and dashboard visualization.

## 10. Exploratory Data Analysis (EDA)

### 10.1 Overview

Exploratory Data Analysis was performed to understand the characteristics of the mutual fund datasets and identify meaningful investment insights.

Several statistical summaries and visualizations were created using Matplotlib and Seaborn.

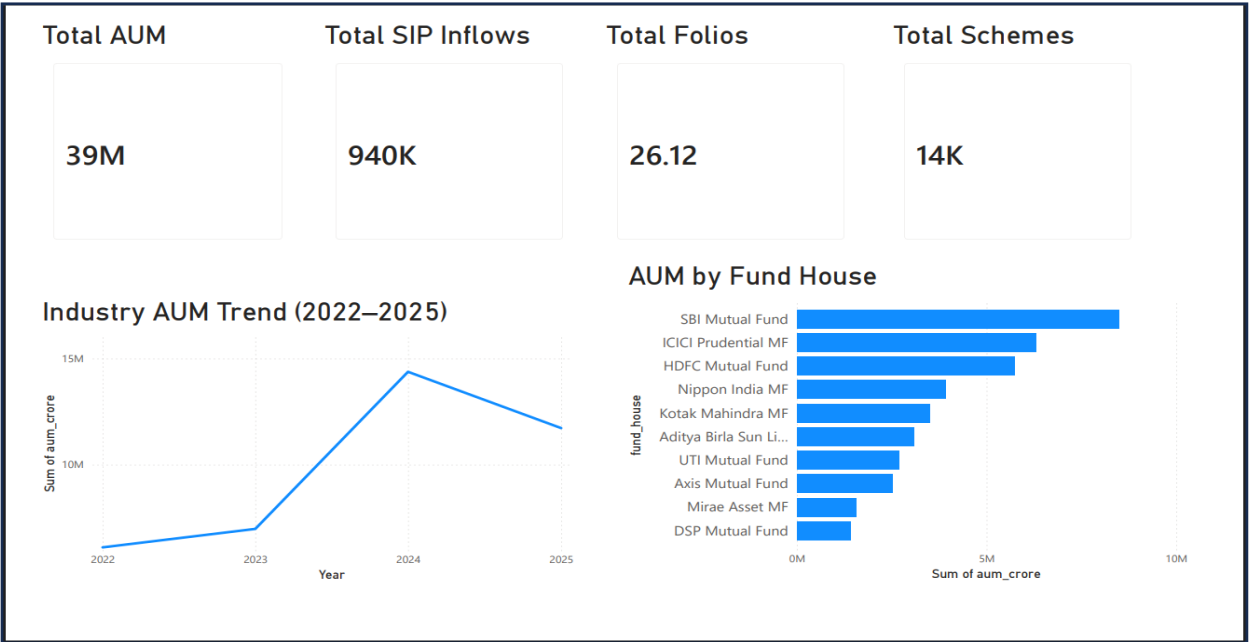
### 10.2 Industry Overview

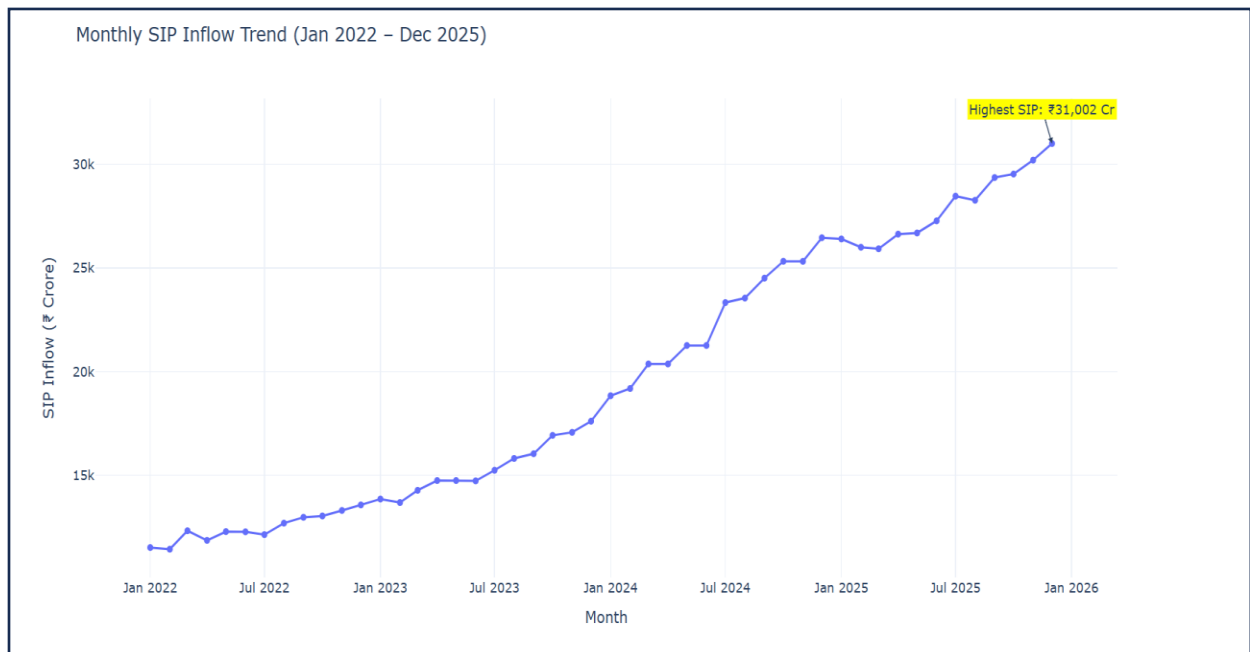
The industry analysis focused on:

- Total Assets Under Management (AUM)
- Monthly SIP Inflows
- Number of Mutual Fund Schemes
- Total Investor Folios

These metrics provide an overall understanding of the mutual fund industry's growth.

*Industry Overview Dashboard*





### 10.3 Investor Analysis

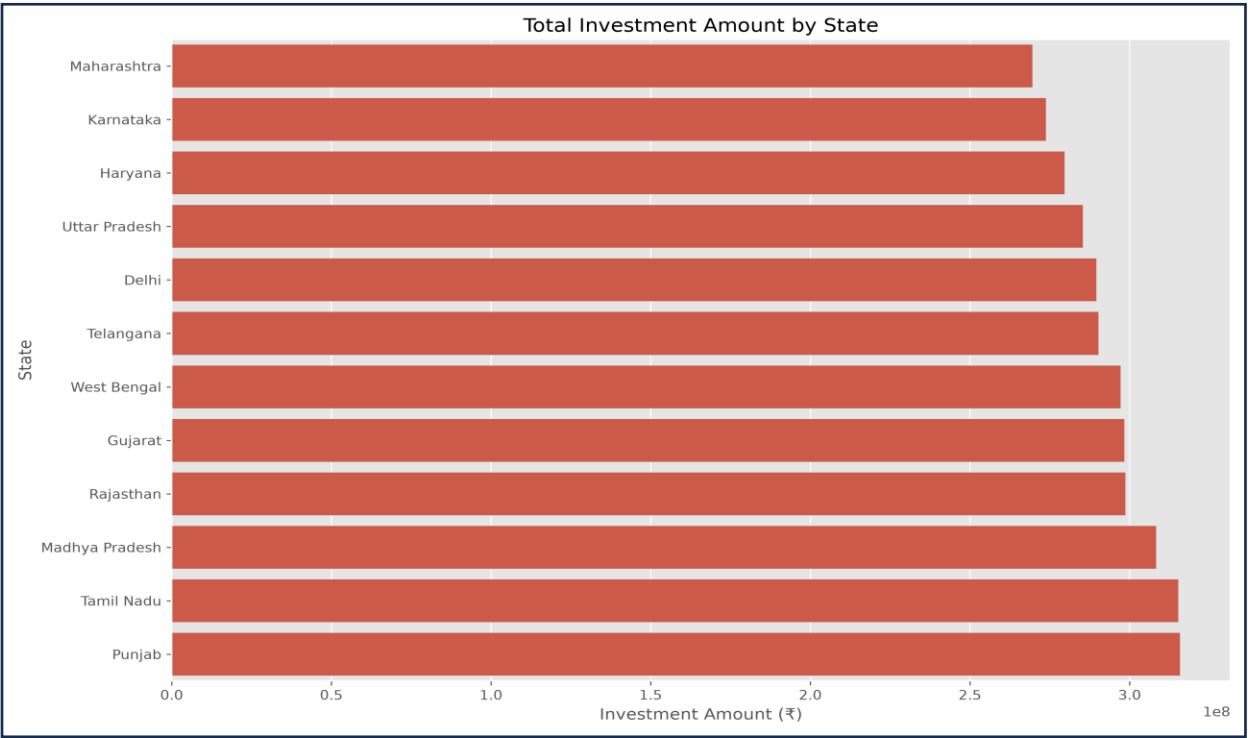
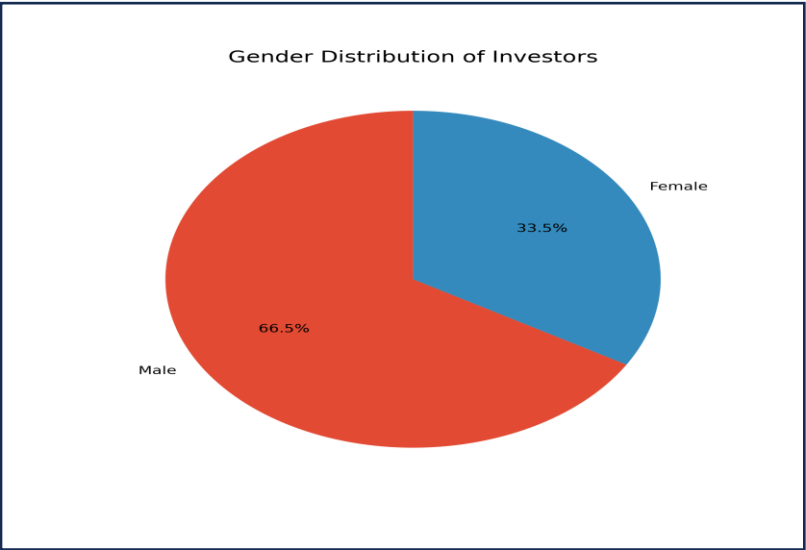
Investor transaction data was analysed to understand investment patterns, demographic distribution, and regional participation in mutual fund investments. Various visualizations were generated to identify trends in investor behaviour and transaction characteristics.

The analysis focused on:

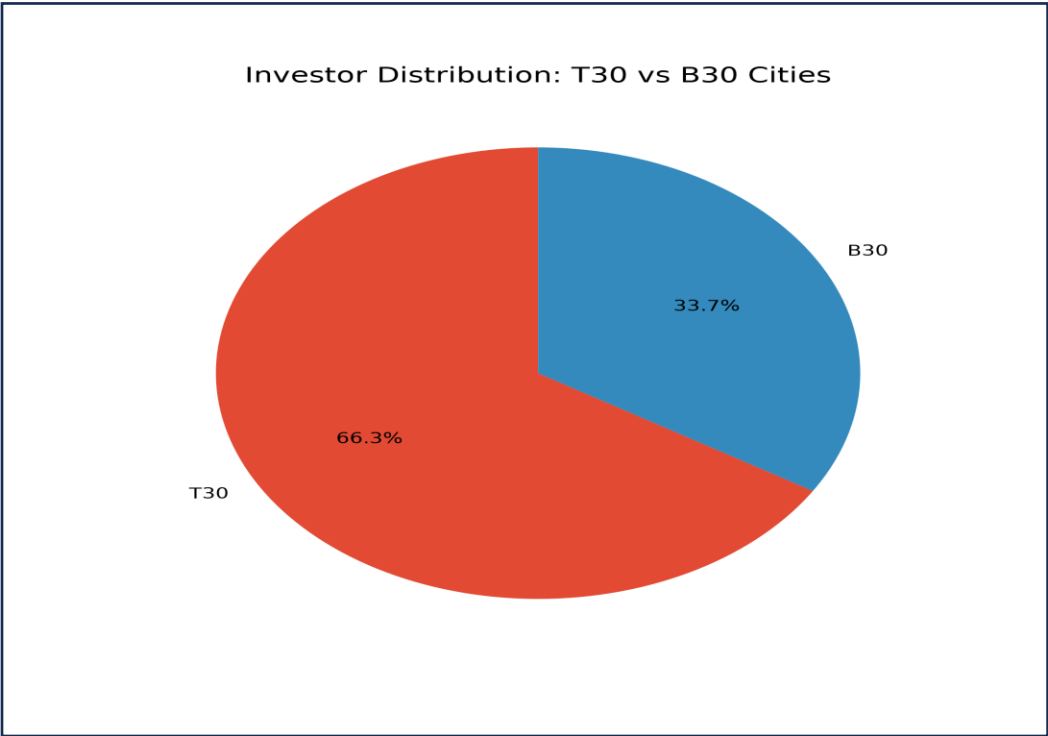
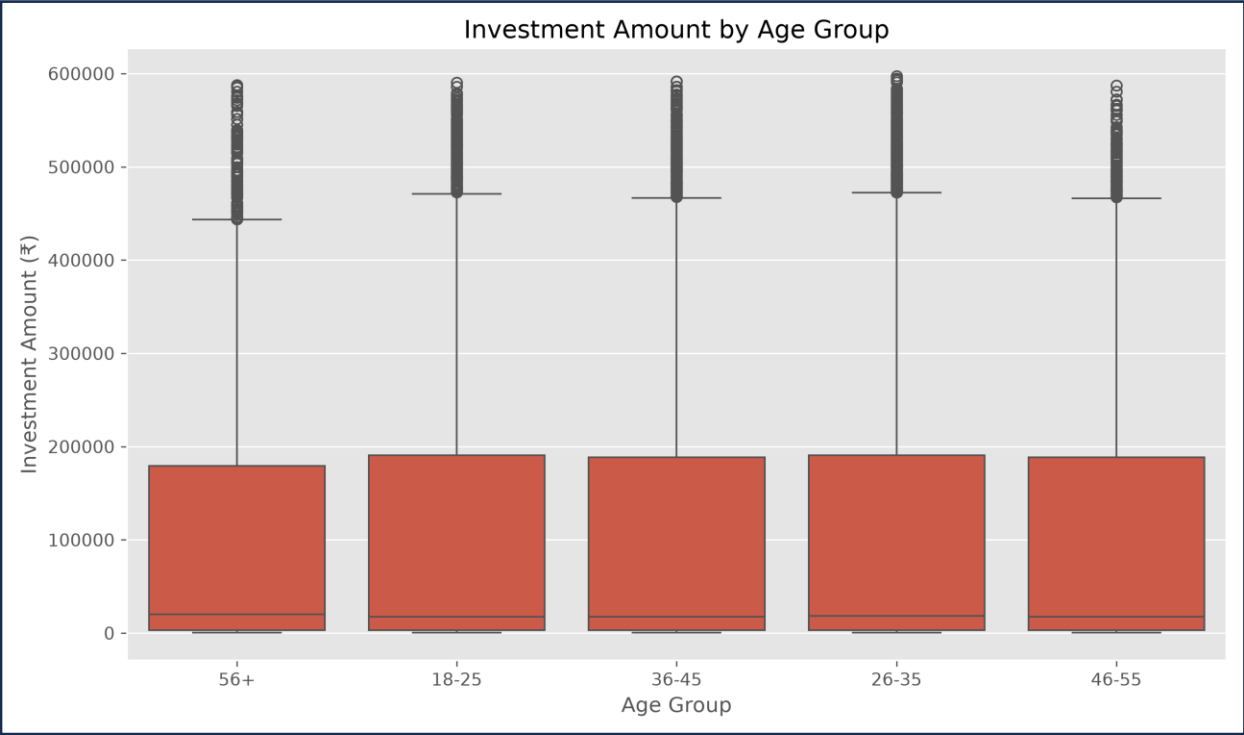
- Gender-wise investor distribution
- State-wise SIP amount
- SIP amount distribution
- T30 vs B30 investor distribution

#### Observations

- Investors from major states contributed a significant share of total SIP investments.
- SIP investment amounts varied across different investors, as shown by the box plot.
- Gender distribution provided insights into investor participation.
- T30 and B30 distribution highlighted investment activity across different geographical regions.







## 10.4 Portfolio Analysis

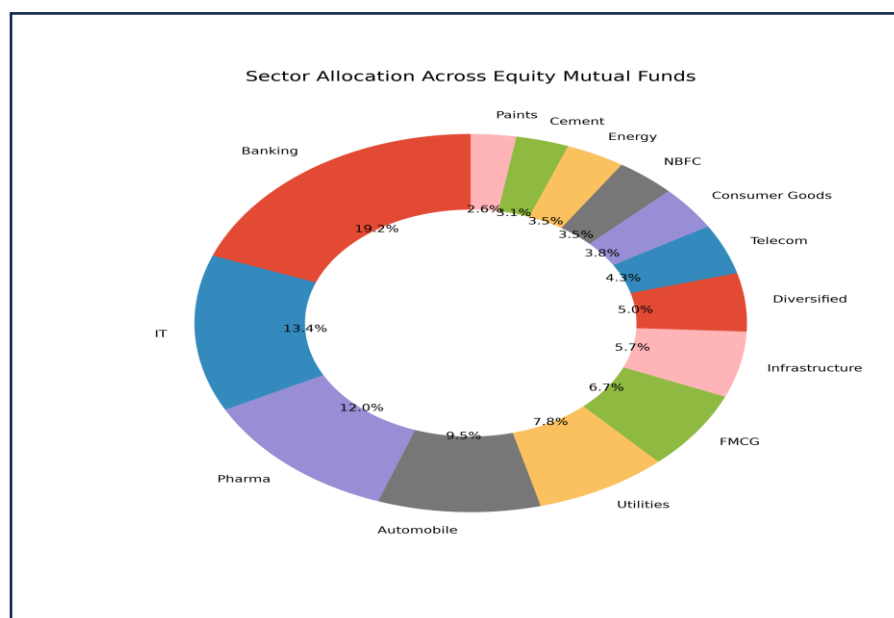
Portfolio allocation analysis was performed to understand how investments were distributed across different industry sectors. Sector-wise diversification provides insights into investment concentration and risk exposure.

### Observations

- Mutual fund investments were diversified across multiple sectors.
- The sector allocation chart highlights the proportion of investments in each industry.
- Diversification helps reduce portfolio risk by spreading investments across sectors.

### ✚ Insert Figure

- **Figure 10.6:** Sector Allocation Donut Chart  
(charts/matplotlib/sector\_allocation\_donut.png)



## 10.5 NAV Trend Analysis

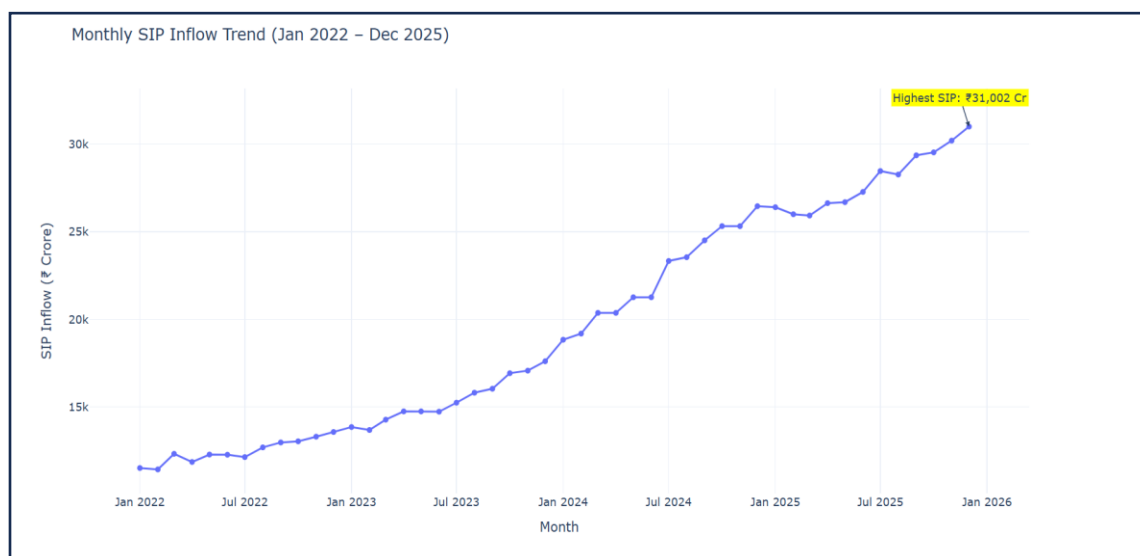
Historical Net Asset Value (NAV) data was analysed to study the performance of mutual fund schemes over time. Trend analysis helps identify long-term growth, market corrections, and recovery periods.

A correlation analysis was also performed to understand the relationship between NAV values and daily returns.

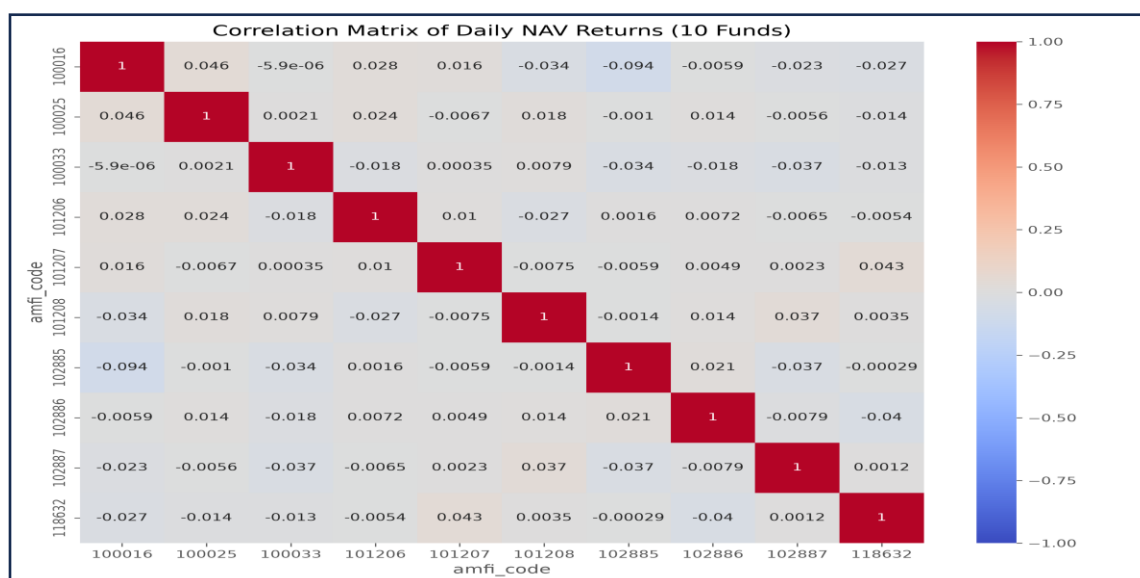
### Observations

- Most mutual fund schemes exhibited steady long-term growth.
- Temporary declines corresponded to periods of market volatility.
- NAV and return analysis provided insights into fund performance consistency.

Figure 10.7: NAV Trend Across All Schemes (*charts/plotly/nav\_trend\_all\_schemes*)



- Figure 10.8: NAV and Return Correlation (*charts/seaborn/nav\_return\_correlation.png*)



# 11. Performance Analytics

## 11.1 Overview

Performance Analytics was conducted to evaluate the historical performance of mutual fund schemes using widely accepted financial metrics. The analysis helps compare funds based on returns, risk, consistency, and downside protection.

The calculations were implemented using Python and Pandas, and the results were exported as CSV reports for dashboard integration and reporting.

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## 11.2 Daily Return Analysis

Daily returns were calculated using historical Net Asset Value (NAV) data.

### Formula

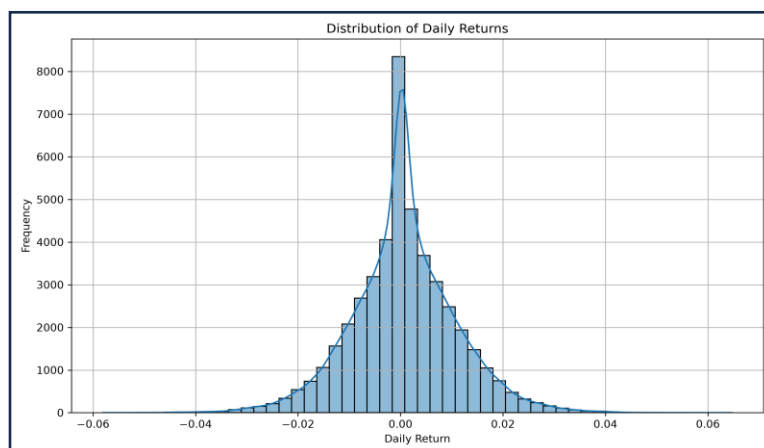
$$\text{Daily Return} = \frac{\text{NAV}_t - \text{NAV}_{t-1}}{\text{NAV}_{t-1}}$$

The daily return values were analysed to understand market volatility and short-term price movements across all mutual fund schemes.

### Observations

- Daily returns fluctuate around zero.
- Most returns lie within a narrow range, indicating moderate day-to-day volatility.
- Extreme positive and negative returns are relatively rare.

### *Daily Return Distribution*



### 11.3 Compound Annual Growth Rate (CAGR)

CAGR was calculated to measure the average annual growth of each mutual fund over the available investment period.

#### Formula

$$CAGR = \left( \frac{NAV_{End}}{NAV_{Start}} \right)^{1/n} - 1$$

where **n** represents the investment period in years.

#### Observations

- Equity-oriented schemes achieved higher long-term CAGR.
- Large-cap and Flexi-cap funds showed relatively stable growth.
- CAGR provided a standardized comparison of long-term fund performance.

#### Top 10 Funds by CAGR

|    | A         | B     | C       |
|----|-----------|-------|---------|
| 1  | amfi_code | years | cagr    |
| 2  | 120505    | 4.4   | 0.32802 |
| 3  | 120505    | 4.4   | 0.32802 |
| 4  | 119598    | 4.4   | 0.32398 |
| 5  | 119598    | 4.4   | 0.32398 |
| 6  | 149324    | 4.4   | 0.32262 |
| 7  | 149324    | 4.4   | 0.32262 |
| 8  | 148569    | 4.4   | 0.31924 |
| 9  | 148569    | 4.4   | 0.31924 |
| 10 | 148567    | 4.4   | 0.3095  |

## 11.4 Sharpe Ratio

The Sharpe Ratio measures risk-adjusted returns by comparing excess returns with total volatility.

Formula

$$Sharpe = \frac{R_p - R_f}{\sigma_p} \times \sqrt{252}$$

where:

- $R_p$  = Portfolio Return
- $R_f$  = Risk-Free Rate (6.5%)
- $\sigma_p$  = Standard Deviation of Returns

Observations

- Higher Sharpe Ratio indicates better returns per unit of risk.
- Funds with consistent performance ranked higher.

*Top 10 Funds by Sharpe Ratio*

|    | A         | B            |  |
|----|-----------|--------------|--|
| 1  | amfi_code | Sharpe_Ratio |  |
| 2  | 148567    | 1.448291     |  |
| 3  | 120843    | 1.306744     |  |
| 4  | 148569    | 1.23493      |  |
| 5  | 119551    | 1.208267     |  |
| 6  | 120505    | 1.180101     |  |
| 7  | 149323    | 1.132122     |  |
| 8  | 100033    | 1.093699     |  |
| 9  | 118632    | 1.081659     |  |
| 10 | 101206    | 1.027213     |  |

## 11.5 Sortino Ratio

The Sortino Ratio improves upon the Sharpe Ratio by considering only downside volatility.

Formula

$$Sortino = \frac{R_p - R_f}{\sigma_d} \times \sqrt{252}$$

where:

- $\sigma_d$  = Downside Standard Deviation

Observations

- Sortino Ratio penalizes only negative returns.
- It is useful for evaluating downside investment risk.

*Top 10 Funds by Sortino Ratio*

|    | A         | B             |
|----|-----------|---------------|
| 1  | amfi_code | Sortino_Ratio |
| 2  | 148567    | 2.385644373   |
| 3  | 120843    | 2.364320032   |
| 4  | 148569    | 2.146913926   |
| 5  | 119551    | 2.140267129   |
| 6  | 120505    | 2.02935294    |
| 7  | 149323    | 1.875101064   |
| 8  | 118632    | 1.850132547   |
| 9  | 100033    | 1.829133962   |
| 10 | 120504    | 1.805293794   |

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## 11.6 Alpha and Beta

Alpha and Beta were calculated using linear regression against benchmark returns.

Alpha

Measures excess return beyond benchmark performance.

## Beta

Measures fund sensitivity to market movement.

### Observations

- Positive Alpha indicates fund outperformance.
- Beta values close to one indicate market-like behaviour.
- Low Beta funds generally exhibit lower volatility.

### *Alpha and Beta Results*

|    | A         | B           | C            |
|----|-----------|-------------|--------------|
| 1  | amfi_code | Alpha       | Beta         |
| 2  | 100016    | 0.037475808 | -0.058268431 |
| 3  | 100025    | 0.04281792  | 0.001158218  |
| 4  | 100033    | 0.271953545 | 0.005103526  |
| 5  | 101206    | 0.213997848 | 0.021086316  |
| 6  | 101207    | 0.108970919 | -0.065289248 |
| 7  | 101208    | 0.060861458 | 0.00026671   |
| 8  | 102885    | 0.170487568 | -0.019486786 |
| 9  | 102886    | 0.028969398 | -0.042125029 |
| 10 | 102887    | 0.162112568 | 0.016682689  |

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## 11.7 Maximum Drawdown

Maximum Drawdown measures the largest decline from a previous NAV peak.

### Formula

$$Drawdown = \frac{NAV}{RunningMax} - 1$$

### Observations

- Lower drawdown indicates better capital protection.
- High-growth funds typically experienced larger drawdowns during market corrections.



## Maximum Drawdown Results

|    | A         | B                | C          |
|----|-----------|------------------|------------|
| 1  | amfi_code | Maximum_Drawdown | Worst_Date |
| 2  | 119599    | -0.525742212     | 10/28/2025 |
| 3  | 119095    | -0.516777537     | 5/11/2026  |
| 4  | 101207    | -0.354469165     | 5/11/2026  |
| 5  | 149324    | -0.311719003     | 1/3/2025   |
| 6  | 119598    | -0.287060057     | 5/14/2025  |
| 7  | 102886    | -0.280011236     | 4/27/2026  |
| 8  | 100016    | -0.247344409     | 9/15/2022  |
| 9  | 120842    | -0.240035113     | 10/17/2024 |
| 10 | 118634    | -0.233448861     | 2/20/2026  |

## 11.8 Fund Scorecard

A composite score was developed by combining multiple financial metrics.

The score considered:

- 3-Year Return
- Sharpe Ratio
- Alpha
- Expense Ratio
- Maximum Drawdown

Each metric was ranked and assigned weights to calculate an overall performance score.

The scorecard enables objective comparison of mutual fund schemes.

### Fund Scorecard

|    | A         | B              | C             | D            | E          | F                | G           | H           | I          | J            | K             | L           |
|----|-----------|----------------|---------------|--------------|------------|------------------|-------------|-------------|------------|--------------|---------------|-------------|
| 1  | amfi_code | return_3yr_pct | expense_ratio | Sharpe_Ratio | Alpha      | Maximum_Drawdown | Return_Rank | Sharpe_Rank | Alpha_Rank | Expense_Rank | Drawdown_Rank | Fund_Score  |
| 2  | 120505    | 18.08          | 1.36          | 1.180101198  | 0.29263583 | -0.181885136     | 8           | 5           | 3          | 15           | 25            | 100         |
| 3  | 120843    | 15.65          | 1.45          | 1.30674397   | 0.27330465 | -0.129739682     | 11          | 2           | 5          | 22           | 13            | 98.14385151 |
| 4  | 100033    | 16.58          | 1.38          | 1.093698744  | 0.27195355 | -0.162172086     | 10          | 7           | 6          | 17           | 20            | 93.03944316 |
| 5  | 119598    | 23.39          | 1.43          | 0.945308258  | 0.30336965 | -0.287060057     | 1           | 14          | 1          | 21           | 36            | 91.87935035 |
| 6  | 148567    | 14.81          | 1.46          | 1.44829117   | 0.26983751 | -0.112657291     | 17          | 1           | 7          | 23           | 8             | 90.71925754 |
| 7  | 149324    | 20.08          | 1.52          | 0.949795825  | 0.30057878 | -0.311719003     | 6           | 13          | 2          | 26.5         | 37            | 80.85846868 |
| 8  | 149323    | 17.16          | 1.61          | 1.132121852  | 0.26598578 | -0.172481058     | 9           | 6           | 8          | 38           | 22            | 78.19025522 |
| 9  | 120504    | 14.41          | 0.8           | 1.026524285  | 0.21194827 | -0.125882756     | 20          | 10          | 13         | 12           | 12            | 76.33410673 |
| 10 | 119094    | 15.18          | 1.38          | 0.998230987  | 0.26076669 | -0.209608839     | 15          | 11          | 9          | 17           | 28            | 74.94199536 |

## 12. Advanced Analytics

### 12.1 Overview

Advanced Analytics extends traditional financial analysis by applying quantitative techniques to estimate investment risk, study investor behaviour, and recommend suitable mutual fund schemes.

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### 12.2 Value at Risk (VaR)

Historical Value at Risk (VaR) was calculated using daily return distributions.

The analysis estimated the maximum expected daily loss at different confidence levels.

#### Results

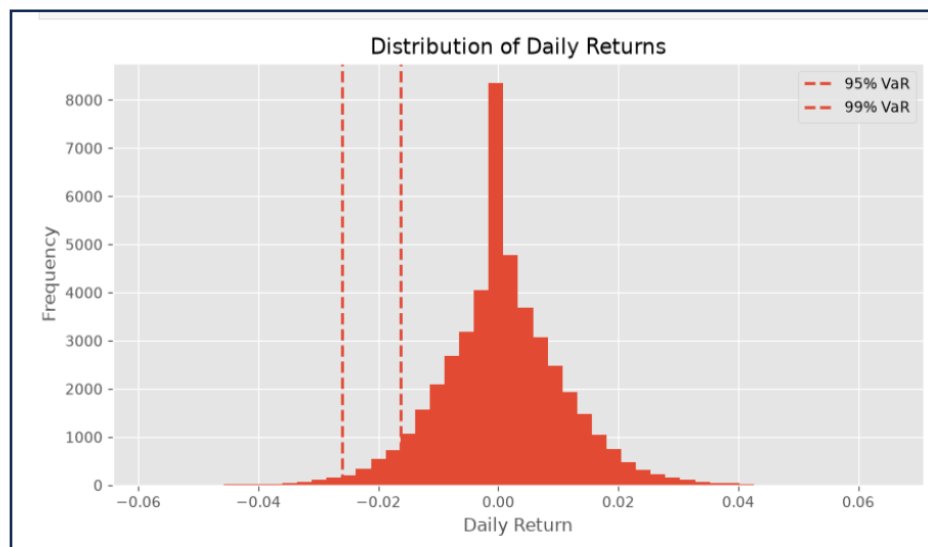
##### Confidence Level Value

95% VaR            -1.6254%

99% VaR            -2.5982%

These values indicate that under normal market conditions, daily losses are expected to remain within these limits at the respective confidence levels.

#### *VaR Histogram*



## 12.3 Cohort Analysis

Investor transactions were grouped based on the month of their first transaction.

A retention matrix was generated to study investor activity over subsequent months.

### Observations

- Investor retention remained high during the initial months.
- Gradual decline in activity was observed over time.
- Cohort analysis provides insights into long-term investor engagement.

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## 12.4 Fund Recommendation System

A rule-based recommendation model was implemented.

Funds were recommended according to investor risk appetite.

### Risk Profile    Recommended Category

Conservative    Liquid, Gilt Funds

Moderate      Large Cap, Flexi Cap

Aggressive    Mid Cap, Small Cap

The recommendation system demonstrates how financial metrics can support investment decision-making.

### *Recommended Funds*

|    | A      | B  | C                                                    | D         | E              | F            | G             | H               | I                  |
|----|--------|----|------------------------------------------------------|-----------|----------------|--------------|---------------|-----------------|--------------------|
| 1  |        |    | scheme_name                                          | category  | return_3yr_pct | sharpe_ratio | expense_ratio | risk_grade      | morningstar_rating |
| 2  | Low    | 14 | ICICI Pru Liquid Fund - Regular - Growth             | Liquid    | 7.68           | 7.68         | 0.74          | Low             | 5                  |
| 3  | Low    | 23 | Kotak Liquid Fund - Regular - Growth                 | Liquid    | 6.18           | 6.18         | 0.6           | Low             | 3                  |
| 4  | Low    | 30 | ABSL Liquid Fund - Regular - Growth                  | Liquid    | 5.14           | 5.14         | 0.79          | Low             | 5                  |
| 5  | Low    | 4  | SBI Magnum Gilt Fund - Regular Plan - Growth         | Gilt      | 6.07           | 1.52         | 0.77          | Low             | 5                  |
| 6  | Low    | 19 | Nippon India Gilt Securities Fund - Regular - Growth | Gilt      | 5.31           | 1.33         | 0.55          | Low             | 4                  |
| 7  | Medium | 22 | Kotak Flexicap Fund - Regular - Growth               | Flexi Cap | 15.65          | 0.98         | 1.45          | Moderately High | 5                  |
| 8  | Medium | 33 | UTI Flexi Cap Fund - Regular - Growth                | Flexi Cap | 15.34          | 0.96         | 1.64          | Moderately High | 5                  |
| 9  | Medium | 5  | HDFC Top 100 Fund - Regular Plan - Growth            | Large Cap | 14.84          | 1.06         | 1.55          | Moderate        | 5                  |
| 10 | Medium | 34 | Mirae Asset Large Cap Fund - Regular - Growth        | Large Cap | 14.81          | 1.06         | 1.46          | Moderate        | 5                  |

## 13. Dashboard Development

### 13.1 Overview

An interactive Power BI dashboard was developed to visualize key financial indicators, mutual fund performance, investor behaviour, and SIP trends.

The dashboard consists of four analytical pages with interactive slicers and drill-down capabilities.

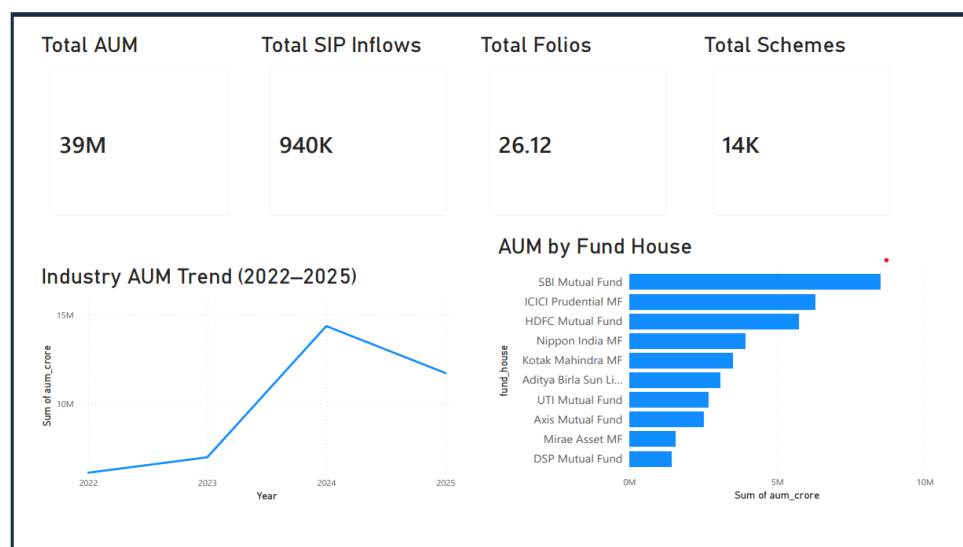
### 13.2 Dashboard Pages

#### Page 1 – Industry Overview

Displays:

- Total AUM
- Total SIP Inflows
- Total Folios
- Total Schemes
- Industry AUM Trend
- AUM by Fund House

#### Dashboard Page 1

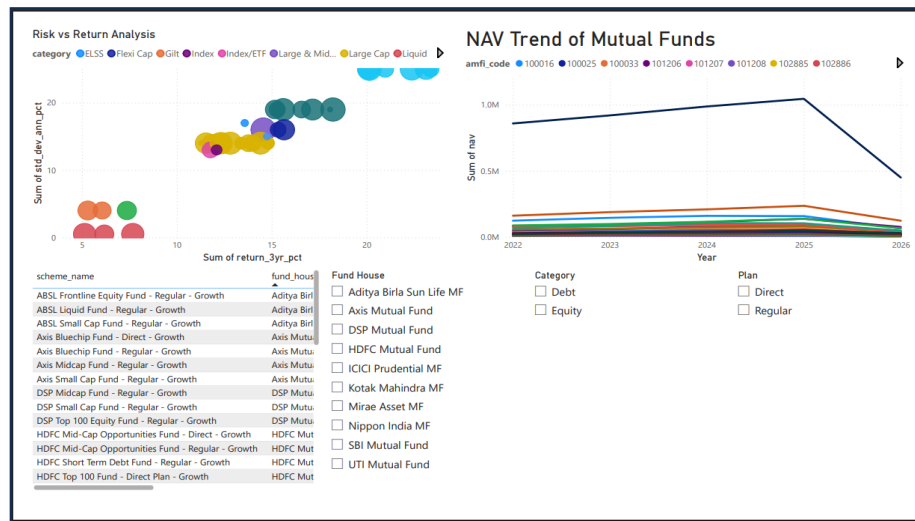


## Page 2 – Fund Performance

Displays:

- Risk vs Return Scatter Plot
- Fund Performance Scorecard
- NAV Trend
- Fund House, Category, and Plan slicers

### Dashboard Page 2

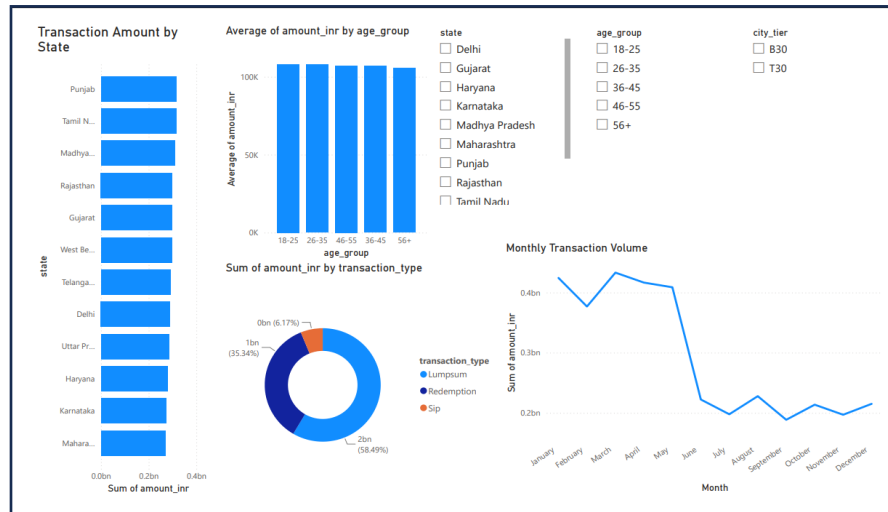


## Page 3 – Investor Analytics

Displays:

- Transaction Amount by State
- Transaction Type Distribution
- Average Investment by Age Group
- Monthly Transaction Volume
- Interactive slicers

## Dashboard Page 3

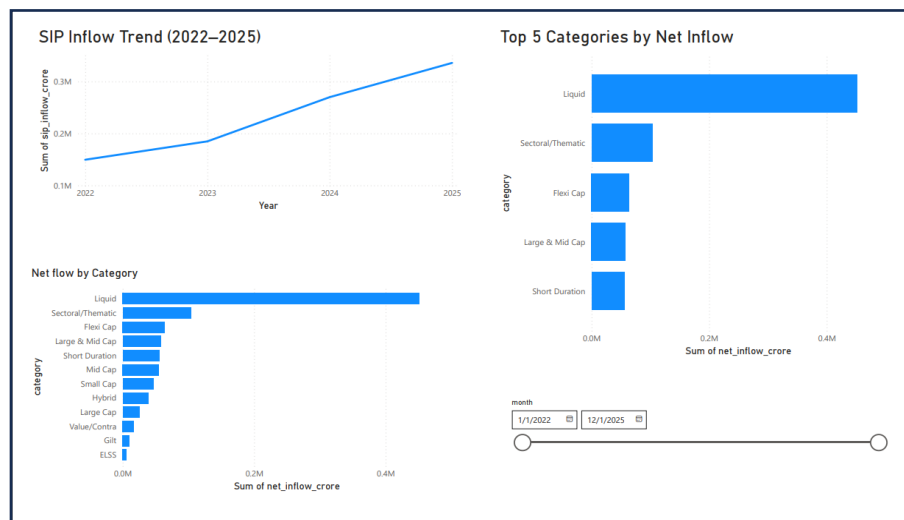


## Page 4 – SIP & Market Trends

Displays:

- SIP Inflow Trend
- Net Inflow by Category
- Top Categories by Net Inflow
- Month slicer

## Dashboard Page 4



## 14. Results and Key Insights

### 14.1 Project Outcomes

The Mutual Fund Analytics project successfully integrated multiple financial datasets into a unified analytics platform. Data was extracted, cleaned, validated, and analysed using Python, SQL, SQLite, and Power BI to generate meaningful investment insights.

The implementation of financial performance metrics, advanced analytics, and interactive dashboards enabled comprehensive analysis of mutual fund performance, investor behaviour, and market trends.

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### 14.2 Key Insights

The following observations were obtained from the analysis:

#### Industry Insights

- The mutual fund industry showed steady growth in Assets Under Management (AUM) and SIP inflows over the analysis period.
  - Equity-oriented mutual funds accounted for the highest share of investor folios.
  - Monthly SIP contributions demonstrated consistent long-term investment behaviour.
- 

#### Investor Behaviour

- SIP was the most frequently used investment method.
  - Investors belonging to the **26–35** and **36–45** age groups contributed a significant portion of investments.
  - Metropolitan and Tier-1 cities generated higher transaction volumes than smaller cities.
  - Verified KYC investors dominated the transaction dataset.
- 

#### Fund Performance

- Several mutual fund schemes achieved strong long-term CAGR values.

- Higher Sharpe and Sortino Ratios indicated better risk-adjusted performance.
  - Alpha values identified funds outperforming benchmark indices.
  - Maximum Drawdown analysis highlighted the importance of downside risk during market corrections.
- 

### **Advanced Analytics**

- Historical Value at Risk (VaR) estimated the maximum expected daily loss under normal market conditions.
  - Cohort Analysis demonstrated gradual reduction in investor retention over time.
  - The rule-based recommendation system successfully classified mutual funds based on investor risk profiles.
- 

### **Dashboard Insights**

The Power BI dashboard enabled users to:

- Monitor industry growth
- Compare mutual fund performance
- Analyse investor demographics
- Explore SIP trends
- Filter results using interactive slicers

The dashboard provides an intuitive interface for financial analysis and business decision-making.



## 15. Challenges Faced

During the implementation of the project, several challenges were encountered.

- Handling large historical NAV datasets efficiently.
- Cleaning inconsistent and missing values from multiple datasets.
- Designing relationships between multiple financial tables.
- Calculating financial metrics accurately using trading-day conventions.
- Developing interactive Power BI dashboards with appropriate relationships and slicers.
- Integrating multiple analytical techniques into a single project.

These challenges were addressed using data preprocessing, validation, modular Python scripts, and systematic testing.

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## 16. Conclusion

The Mutual Fund Analytics project successfully demonstrated the complete lifecycle of a financial data analytics application.

The project integrated multiple datasets and transformed them into meaningful business insights using Python, SQL, SQLite, and Power BI. Comprehensive exploratory data analysis, performance analytics, advanced financial analytics, and interactive dashboards were developed to evaluate mutual fund performance and investor behaviour.

The implementation of financial metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, and Value at Risk enhanced the analytical capability of the system. The Cohort Analysis and Fund Recommendation System further demonstrated practical applications of data analytics in financial decision-making.

Overall, the project provided valuable hands-on experience in data engineering, financial analytics, business intelligence, and software development while addressing a real-world investment analysis problem.

## 17. Future Scope

The current project can be extended with additional features to improve its analytical capabilities.

Future enhancements include:

- Integration of live mutual fund data using APIs.
- Machine learning models for mutual fund return prediction.
- Portfolio optimization using Modern Portfolio Theory.
- Monte Carlo simulation for future NAV forecasting.
- Automated ETL scheduling using cron jobs or Task Scheduler.
- Streamlit-based web application for online access.
- Automated email reporting and notification system.
- Real-time dashboard updates.

These enhancements would make the platform more scalable, intelligent, and suitable for production environments.

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## 18. References

1. Association of Mutual Funds in India (AMFI)
2. Reserve Bank of India (RBI)
3. National Stock Exchange (NSE)
4. Python Documentation
5. Pandas Documentation
6. NumPy Documentation
7. Matplotlib Documentation
8. Seaborn Documentation
9. SQLite Documentation
10. Microsoft Power BI Documentation
11. Bluestock Fintech Internship Datasets

## **Appendix A – GitHub Repository**

Include:

GitHub Repository:

<https://github.com/Harshitha-0510/MutualFundAnalytics>

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## **Appendix B – Dashboard Screenshots**

Include one screenshot each of:

- Industry Overview
- Fund Performance
- Investor Analytics
- SIP & Market Trends